

Rising Waters: AI-Powered Flood Prediction System

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PROJECT DESCRIPTION

Rising Waters harnesses deep learning, satellite remote sensing imagery, and real-time sensor streams to provide highly accurate, localized flood forecasting and public safety warning infrastructure. The platform features an advanced **Flood Prediction module** utilizing a custom PyTorch UNet convolutional neural network. This model digests historical rainfall datasets and multi-band Sentinel-2 satellite imagery to generate high-resolution surface inundation projections.

The backend engine is engineered on top of **FastAPI** and integrates spatial analytics powered by a **PostgreSQL / PostGIS database**. Real-time field observations are ingested into a live **Telemetry Dashboard** that coordinates physical IoT water-level sensor flows. Additionally, an **Automated Alert Dispatcher** is configured to trigger immediate, targeted SMS warning alerts via Twilio integrations to at-risk local residents and municipal defense agencies when predicted thresholds are breached. Built with scalability in mind, the platform uses containerized environments managed through Docker, securing production variables using `.env` files to help municipalities protect lives and build long-term disaster resilience.

SCENARIOS

Scenario 1 — Preemptive Inundation Mapping (Satellite Analysis / UNet Model)

An environmental risk manager monitors an incoming monsoon system. They upload the latest optical and infrared Sentinel-2 satellite passes of the local river basin to the dashboard. The UNet deep learning model instantly processes the bands, generating a high-precision surface water map within seconds that outlines low-lying neighborhoods most at risk of inundation.

Scenario 2 — Real-Time IoT Triggered Evacuation (Sensor Stream / SMS Alerts)

During heavy rainfall, upstream physical IoT sensors record rapid, anomalous increases in water flow rates and water depth. As these real-time streams cross critical alert thresholds, the central system automatically triggers the alert subsystem, broadcasting localized, geo-targeted emergency SMS warnings through Twilio to citizens in immediate downstream flood paths.

Scenario 3 — Urban Planning & Long-Term Simulations (Historical Analysis / Planning)

A municipal planning committee evaluates potential land developments along a flood-prone plain. By inputting historical flood records and simulating virtual heavy precipitation events, the platform models structural vulnerability patterns, helping planners design resilient drainage pathways and choose safe development zones.